

8(a)	<i>any 4 of:</i> 1 (presynaptic) membrane depolarised ; 2 calcium (ion) <u>channels</u> open / (presynaptic membrane becomes) more permeable to Ca^{2+} ; 3 Ca^{2+} move in (to presynaptic, neurone / knob) ; R membrane 4 causes vesicles (of ACh) to move towards (presynaptic) membrane ; 5 vesicle fuses with (presynaptic) membrane / exocytosis ;	4
8(b)	<i>any 2 of:</i> 1 no Na^+ enter (postsynaptic neurone) ; 2 no depolarisation of (postsynaptic) membrane ; 3 no, action potentials / impulses (transmitted) ;	2
8(c)(i)	A – myelin sheath / Schwann cell ; B – sarcolemma ; C – mitochondria ;	3
8(c)(ii)	<i>any 2 of:</i> 1 insulates axon / prevents movement of ions / AW ; 2 increases speed (of impulses) ; 3 saltatory conduction / described ;	2

Question	Answer	Marks
8(c)(iii)	1 produces / releases, ATP ; <i>any 3 of mp2-5:</i> 2 (for) production / recycling, of ACh ; 3 (for) production / movement, of vesicles or exocytosis ; 4 (for) contraction of sarcomere ; 5 (for) sodium potassium pumps / active transport ;	4